VF Asia Footwear Laboratory Testing Manual

Test Method for Flexing Resistance of Footwear or Outsole - China(GB) Standard	Test method # VF- G05 Page 1 of 2
Date Issued : January 09 , 2013 Revision # :	Created by : Nila Babia Erazo / Chris Adam

PURPOSE AND SCOPE: To determine the resistance of completed footwear of outsole or just outsole which is flexed at 50 degree angle. After the predetermined flexes, the footwear is measured for crack length and split length at the outsole and welt.

REFERENCES: GB/T 3903.1

EQUIPMENT AND MATERIAL:

1. Flexing resistance machine –China (GB) Standard
2. Folding last - different sizes
3. Chisel or piercing tool.
4. Vernier caliper (graduated to 0.02 mm)
5. Pen
6. Footwear or outsole.

SAMPLE PREPARATION:

1. Condition the test sample in a standard controlled environment at 23 +/- 2 °C and at 55 +/- 5 % RH for at least 48 hours to allow the cement to crosslink completely.

2. Place the folding last to the shoe (sandal or outsole). The forepart of the folding last should not fit against the upper of the footwear. The folding last is shorter by 5 mm or half size. If the fitting is still too tight, use folding last which is 1 size lower.

3. Check the folding area of the last and draw a line across the outsole .This line is regarded as the flexing line of the sample.

4. Check the width of the line and mark the medial point. Position the test sample to the medial point such that the chisel tip is located on the marked line.

5. Push the chisel tip just far enough into the test sample so that a 5mm wide slit is produced throughout the entire thickness of the sample.

6. So that the crack can be seen easily, for colored materials, mark with silver pen the 5 mm slit while for light colored materials, mark with regular color pen.

REPETITIONS: Either one or both of a pair of completed whole footwear or outsole.

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TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Lace and tie the footwear or outsole in a normal manner. (Lace opening should typically be about 3cm).

2. Clockwise rotate the handle on the front clamp until the forepart of the specimen can be placed between the top and bottom jaws.

3. Fit the forepart of the footwear or outsole to the front clamp is such a way that its outsole faces upwards and flexing position of the metatarsus coincides with the moveable axle of the clamp. Counterclockwise rotate the handle until the front clamp holds the forepart of the shoe.

4. Rotate the handle wheel and the knob until the forepart of the specimen is locked.

5. Remove the set screw to allow the back clamp to move free. Fit the backend of the footwear or outsole to the back clamp and then lock the back end of the footwear or outsole in such a way that the set bolts are wound with the shoelaces.

6. Set the number of flexes and test frequency on the control panel in accordance with the test requirements.

7. Initiate testing

8. Check the footwear or outsole at least every 5,000 cycles over the first 25,000 cycles, after 25,000 cycles, check the shoes at least every 10,000 cycles.

9. At the end of the desired cycles, measure the crack length

REPORT: Length of crack in mm.

Note: In the report include other failures such as delamination between midsole to outsole, delamination in the flexed area, cracking of components, etc.